

### 3 ADDRESSING CHARGING NEEDS FOR INDIA'S MOST ADOPTED EV SEGMENT

While innovations in products and solutions keep a startup a step ahead of the competition, innovation in the business model ultimately decides the runway and scale. Such is the story of IIT Madras-incubated battery swap solutions provider Esmito Solutions

**W**hile the battery swap story in India has seen its ups and downs, it is pointed to as the most viable solution for charging needs in the electric two- and three-wheeler segments, especially in the B2B sector, i.e., logistics, transportation, and allied segments.

In a nutshell, every electric vehicle (EV) form factor, plying on intercity routes, has become a successful use case for battery swapping. This is because it allows these vehicles to spend more time on roads and less time getting recharged.

Incubated in 2018 at IIT Madras, Esmito is focusing on the B2B side of electric two- and three-wheelers. Dr Prabhjot Kaur, Co-founder and Co-CEO of Esmito Solutions, told Electronics For You, "We have scaled up in serving swapping solutions with several players in India now and are starting to conduct pilots outside India."

Apart from these international locations where the tech developed by Esmito in India has become one of the first to be tested, the startup is fully operational in Mumbai in the E3W and E2W segment and has recently started its operations in Delhi and Hyderabad.

Dr Kaur was previously CEO of the Centre for Battery Engineering and EVs at IIT Madras, where she was instrumental in several technology translations and licensing EV technologies to companies for production. Notably, IIT Madras



*Dr Prabhjot Kaur, Co-founder and Co-CEO of Esmito Solutions*

also happens to be the ground zero of Professor Ashok Jhunjhunwala, who is regarded as one of India's top voices advocating battery swap technology.

The startup operates on two business models, which include the CAPEX approach, where it sets up and sells these stations to third parties, and the OPEX approach, where it sets up and operates battery swap stations on its own. Only about 20% of the total swap stations it has deployed so far are on the latter model.

Revenue generation for the startup also takes place from the annual maintenance contracts it signs with most of the parties to which it sells these swap stations. Moreover,

there are also charges for software used to run these stations.

While it offers to set up battery swap stations in configurations of up to 20 batteries, the majority of its clients opt for the six-battery configuration setup.

Apart from the battery swap stations business, the startup also offers battery management systems and energy-as-a-service as its products and solutions, which encompass CAN-enabled smart batteries for electric two-wheelers and electric three-wheelers. These smart batteries are IoT-enabled, and their performance characteristics can be monitored remotely. It also retails standalone 48V, 60V, and 72V battery packs and standalone swap stations.

The startup's battery manufacturing facility in Navi Mumbai can manufacture 2500 batteries (2kWh each) each month, and its Hyderabad-based swap station facility can manufacture 100+ swap stations a month. Having raised ₹140 million, it says it is in the middle of fundraising rounds for around \$15 million.

Hasan Ali, Co-CEO and Co-founder of Esmito Solutions, was previously associated with Amazon as a senior leader working towards launching and scaling large scale fulfilment operations. He has also worked with Yokohama as the Head of Fulfilment, and with Cummins in its business strategy and program management vertical. **EFY**

—Mukul Yudhveer Singh